Next Billion AI Index

Ambrish Rawat

/ambrishrawat @iambrishing

"Within our community, there is no consensus on what (AI) works for us"

AI Developer, Kenya – formative study participant

Not asking which model scores highest on a benchmark.

Asking: given my context, my constraints, my users — which AI technologies serve me?

Performance matters. But it’s not the whole story.

What benchmarks ask

How capable is this model at peak, in ideal conditions?

- Reasoning and Math
- Code generation
- Safety red-teaming
- Multilingual NLP scores
- Latency and throughput

What a next-billion developer asks

Can I deploy this where my users actually live?

- Can I afford this at scale?
- Will it work offline? On low bandwidth?
- Can I adapt it to my language or domain?
- Will my users trust it?
- Can my community govern it?

Adjacent efforts — but silent on utility, adoption, and next-billion context.

HELM / METR

“How capable is the model?”

Holistic evaluations across reasoning, language, safety and agentic task horizons.

Foundation Model Transparency Index

“How transparent is the developer?”

Structured rubric for documentation, data, compute, and governance disclosure.

Artificial Analysis Openness Index

“How open is the model?”

Rates open weights, code, data and licensing across model families.

AI Index

“How is AI progressing globally?”

Tracks research output, investment, and capability trends over time. Broad ecosystem lens.

Global Index on Responsible AI

"Are responsible AI targets implemented?"

Comparative benchmark on responsible AI practices Governance focused, country-level.

Government AI Readiness Index

“Is this country ready to deploy AI?”

Evaluates national infrastructure, talent, and regulatory capacity. Top-down, policy-centric.

Next Billion AI Index

Effective Efficiency

Can the system deliver maximum value per unit cost and resource?

Operational Practicality

Can the system be adapted and sustained in heterogenous real-world settings?

Societal Integrity

Does the system respect, represent, and include local values and does it allow for shared stewardship?

Nexbax · 3 themes · 10 dimensions · 1 diagnostic framework

A

Effective Efficiency

A1 · Cost Effectiveness and Price Performance

A2 · Resource efficiency

B

Operational Practicality

B1 · Adaptability and Customizability

B2 · Interoperability

B3 · Resilience, Reliability and Robustness

B4 · Usability and Automation

B5 · Education and Empowerment

C

Societal Integrity

C1 · Trustworthiness and Ethics

C2 · Multiculturalism, Inclusivity and Pluralism

C3 · Openness and Collaboration

What does it look like to use *nexbax* in practice?

1 Define rubrics per dimension

Each dimension gets a ladder, e.g. weak, moderate, strong or feasible, potential, proven. Applied to developer facing system properties in this first version.

Weak	e.g., no fine-tuning, cloud-only, English defaults
Moderate	e.g., Some adapters, major languages, basic safety docs
Strong	e.g., Modular, edge-viable, community governance

2 Build on existing benchmarks

Where they exist, established benchmarks inform dimensions. Where they don't, proxy evidence and practitioner judgment fill the gap.

Resource efficiency	MLPerf, AI Energy Score, Intelligence per Watt
Multiculturalism and inclusivity	FLORES-200, MMTEB, low-resource benchmarks
Trustworthiness	MLCommons AILuminate, DecodingTrust
Adaptability	Proxy: fine-tuning APIs, adapter ecosystems

3 Co-design with local stakeholders

For the global majority, usefulness is context-dependent. Rubrics must be defined and adapted with local communities.

- Choose which dimensions matter most for your use case
- Validate what “strong” means in your deployment context
- Surface trade-offs that are invisible from the outside
- Develop a vocabulary to reason together

How model class shapes the trade-offs?

	Effective Efficiency	Operational Practicality	Societal Integrity
Closed API models Claude, GPT. Gemini	Per-token cost accumulates fast for large user bases	Easy to deploy but no fine-tuning, often cloud only	Opaque governance, Western cultural defaults
Open-weight models Granite, Qwen, Mistral	No per-token cost, edge/offline viable	Highly adaptable but requires skilled team	Community governance possible; but not automatic
Open-weight + orchestration LangChain, Mellea	Added abstraction cost; coding agents succeed, chatbots less so	Best interop story, modular components	Openness depends on base model; complexity obscures accountability

Characteristic gap

Trade-off present

Characteristic strength

We tested *nexbax* with those building for next-billion markets.

Who

Founders / CTOs	4
Developers / Engineers	4
Product / Analytics leads	3

Where

- India
- Kenya
- Ghana
- United States

“All the tools we use to evaluate vendors beyond localisation — you hit all of them. Ease of use, reliability, cost, setup time.”

— Product manager, India

Top considerations cited

Cost-effectiveness	10/11
Usability & automation	5/11
Trust & ethics	4/11

How they would use it

Eliminate
Rule out technologies that fail on critical dimensions for their context

Choose
Select the right model for a specific deployment

Synthesise
A digest — research done, options surfaced, context preserved

Use cases: Visual analytics · Content authentication · Education · Customer service · AgTech · Information technology

“Isn’t this just another index?”

1 Indices move incentives

Stanford FMTI demonstrated this — once transparency was measured systematically, AI companies improved. Nexbax wants the same effect for deployability and inclusivity.

2 Nexbax is not a universal score of usefulness; it’s a first-stage diagnostic

It surfaces trade-offs. The right model for a coding agent in a well-connected enterprise is not the right model for a multilingual advisory service in India. Nexbax makes that legible.

3 In the spirit of the UN SDGs

The SDGs didn't rank countries and declare a winner. They made the dimensions of human development visible, discussable, and actionable. That's the mode we're operating in.

What does *nexbax* mean for Sovereign AI?

India

Application-specific AI — not frontier model supremacy.

Small models tailored to sector needs. Open-weight platforms as the path to achieve more with less.

Economic Survey 2025–26, Ch. 14

Africa

Local talent and ecosystems paramount. Governance is a first-order adoption condition.

AI foremost to address African challenges — healthcare, food security, clean energy. Serving the continent's needs and priorities.

AU Continental AI Strategy, 2024

Locally hosted ≠ locally useful

A country can host its own compute and still build a system nobody uses — too costly, too brittle, too culturally narrow to earn trust or adoption.

Sovereignty requires deployability

Systems must be affordable, resilient, and governable in local context — not just technically sovereign at the infrastructure layer.

Nexbax operationalises the question

A shared vocabulary for developers and policymakers to ask: does this system create the conditions for useful, inclusive, and locally governed AI?

Generative Computing is *nexbax*-aligned by design

Paradigm for building reliable, composable AI systems which treat LLMs as programmable components in a software stack with typed interfaces, composable modules and explicit boundaries.

SLMs → Resource Efficiency

A1–A2 · Cost and Resource Efficiency

Granite 4.1: 3B–8B models matching or exceeding much larger predecessors. Apache 2.0, 12 languages, deployable at the edge. aLoRAs carry KV state across steps and avoid re-prefill between tasks.

Modularity → Adaptability

B1 · Adaptability and Customizability

Granite Libraries: adapter functions for RAG, safety, and core capabilities. Each is independently trained, composable, replaceable — like software dependencies. Small models reach large-model accuracy on narrow tasks.

Open + Structured → Integrity

C3 · Openness and Collaboration

All models, libraries, and Switch toolkit released open-source. Typed interfaces and explicit adapter boundaries make behavior reviewable, governable, and localizable (not buried in prompts).

Mellea

mellea.ai

Open-source library for generative computing. Calls adapter functions as typed Python functions — turning unpredictable text generation into deterministic, composable software.

Project Granite Switch

github.com/generative-computing/granite-switch

Compose Granite adapter functions into a deployable model in minutes. Small models, dynamically switched capabilities — no full retraining required.

AI Progress can’t be measured by benchmark supremacy alone.

It must be measured by whether systems can be deployed, adapted, trusted, and sustained where the next billion users live and work.

nexbax.github.io

arxiv.org/abs/2606.00359